Climate and Coral Reefs

Coral reefs cover less than one percent of the ocean floor, yet they support about a quarter of all marine species. Reef-building corals live in partnership with tiny algae that provide most of their food through photosynthesis.

When ocean water stays too warm for too long, corals expel these algae and turn white. This process is called coral bleaching. Bleached corals are not dead, but they are starving and far more likely to die if the heat continues.

Scientists monitor reefs in the Arabian Gulf closely because these corals already survive some of the hottest seas on Earth. Studying how Gulf corals tolerate heat may help protect reefs elsewhere as oceans warm.

Protecting reefs

Marine protected areas limit fishing and anchoring so that damaged reefs can recover. Coral nurseries grow fragments of heat-tolerant coral and replant them on degraded reefs. Reducing carbon emissions remains the most important long-term protection for coral reefs.